

Exercise 2 SQL Fundamentals

1. Selecting & Sorting

① SELECT (*)
FROM employees;

employees

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000 50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000 72000	2017-04-30	Johannesburg

② SELECT DISTINCT department
FROM employees;

employees

Department
IT
HR
Finance
Marketing

③ SELECT first_name, last_name
FROM employees
ORDER BY salary DESC

employees

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

④ SELECT id, first_name, last_name, salary
 FROM employees
 ORDER BY salary DESC
 Limit 3

employees

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

⑤ Select Id, first_name, last_name, department
 FROM employees
 WHERE department = 'IT';

employees

id	first_name	last_name	Department
5	Eva	Naidoo	IT
1	Alice	Green	IT

⑥ SELECT id, first_name, last_name, department, salary
 FROM employees
 WHERE department = 'finance' AND salary > 60000

employees

id	first_name	last_name	department
5	Eva	Naidoo	IT
1	Alice	Green	IT

id	first_name	last_name	department	salary
3	Cathy	Zulu	finance	65000

⑦ SELECT id, first_name, last_name, department
 FROM employees
~~WHERE department = 'HR' OR 'Marketing';~~
 WHERE department IN (~~'HR', 'IT'~~) ('HR', 'Marketing');

employees

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

⑧ SELECT id, first_name, last_name, department;
 FROM employees;
~~WHERE NOT IN IT~~
 WHERE employees NOT IN IT;

employees

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

⑨ SELECT id, first_name, last_name, department
 FROM employees
 WHERE department IN ('IT', 'HR', 'Finance');

employee

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

⑩ SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE ~~Salary~~ department = 'IT' AND salary > 65000 AND
city = 'Johannesburg';

Employees

id	first_name	last_name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Nardoo	IT	72000	Johannesburg